CTF Learning Report

1. Which challenge was easiest, and why?

The easiest challenge was finding hidden information using basic Linux commands. It was simple because commands like `ls`, `pwd`, and `cat` helped me explore files and identify useful information.

2. Which challenge was hardest, and what made it hard?

The hardest challenge was identifying hidden services and understanding network information. It was challenging because I needed to analyze the results carefully and use the correct commands to find clues.

3. Which Linux commands did you use during the CTF?

I used the following Linux commands:

- `ls` – List files and directories.
- `cd` – Change directories.
- `pwd` – Display the current directory.
- `cat` – Read file contents.
- `find` – Search for files.
- `grep` – Search for specific text.
- `chmod` – Change file permissions.
- `whoami` – Display the current username.
- `nslookup` – Check DNS information.
- `nmap` – Scan open ports and services.

4. What did you learn about permissions?

I learned that Linux permissions control who can read, write, and execute files. I also understood the importance of `r`, `w`, and `x` permissions and how the `chmod` command is used to modify them.

5. What did you learn about DNS?

I learned that DNS (Domain Name System) converts domain names into IP addresses. Tools like `nslookup` help identify DNS records and understand how domain name resolution works.

6. What did you learn about ports?

I learned that ports help identify different services running on a system. Common ports include:

- Port 21 – FTP
- Port 22 – SSH

- Port 53 – DNS
- Port 80 – HTTP
- Port 443 – HTTPS

7. What did you learn from reading Nmap output?

I learned how to identify open, closed, and filtered ports, discover running services, and understand service versions. Nmap results help identify potential security risks and understand the target system's network exposure.

8. What would you revise before the next CTF?

Before the next CTF, I would revise Linux commands, file permissions, DNS concepts, common network ports, Nmap scanning techniques, and basic enumeration methods. I would also practice more hands-on CTF challenges to improve my problem-solving skills.

Challenge 6 – Wireshark Packet Detective (30 Points)

1. Which protocol resolves the domain?

DNS (Domain Name System) resolves the domain name `example.com` into an IP address.

2. Which destination port is used for the web connection?

Port 443 is used for the HTTPS web connection.

3. Which transport protocol establishes the connection?

TCP (Transmission Control Protocol) establishes the connection.

4. Which packets represent the TCP three-way handshake?

The TCP three-way handshake consists of:

- Packet 3 – SYN
- Packet 4 – SYN, ACK
- Packet 5 – ACK

5. Which protocol appears after the TCP handshake?

TLS (Transport Layer Security) appears after the TCP handshake. The client sends a TLS Client Hello.

Final Flag

Correct flag format

JVL{DNS_TCP_443_TLS}

Conclusion: This challenge helped me understand DNS resolution, TCP three-way handshakes, HTTPS port 443, and TLS communication using Wireshark packet analysis.